

28. For $x \geq y > 1$, let $\log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right) = k$, then

the value of k can never be equal to

- (a) -1 (b) $-\frac{1}{2}$ (c) 0 (d) 1

29. If $A = \begin{bmatrix} \sin 2\theta & 2\sin^2 \theta - 1 & 0 \\ \cos 2\theta & 2\sin \theta \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then which of

the following statements is/are correct?

1. $A^{-1} = \text{adj } A$
2. A is skew-symmetric matrix
3. $A^{-1} = A^T$

Select the correct answer using the code given below:

- (a) 1 only (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

30. What is the coefficient of x^{10} in the expansion of $(1-x^2)^{20} \left(2-x^2 - \frac{1}{x^2} \right)^{-5}$?

- (a) -1
(b) 1
(c) 10
(d) Coefficient of x^{10} does not exist

31. If the 4th term in the expansion of $\left(mx + \frac{1}{x} \right)^n$ is $\frac{5}{2}$, then what is the value of mn ?

- (a) -3 (b) 3 (c) 6 (d) 12

32. If a, b and c ($a > 0, c > 0$) are in GP, then consider the following in respect of the equation $ax^2 + bx + c = 0$;

1. The equation has imaginary roots.
2. The ratio of the roots of the equation is 1: ω where ω is a cube root of unity.
3. The product of roots of the equation is $\left(\frac{b^2}{a^2} \right)$.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

33. If $x^2 + mx + n$ is an integer form all integral values of x , then which of the following is/are correct?

1. m must be an integer
2. n must be an integer

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

34. In a binomial expansion of $(x+y)^{2n+1}(x-y)^{2n+1}$, the sum of middle terms is zero. What is the value of $\left(\frac{x^2}{y^2} \right)$?

- (a) 1 (b) 2 (c) 4 (d) 8

35. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7\}$. What is the number of onto functions from A to B ?

- (a) 10 (b) 20 (c) 30 (d) 32

36. What is $\frac{\sqrt{3} \cos 10^\circ - \sin 10^\circ}{\sin 25^\circ \cos 25^\circ}$ equal to?

- (a) 1 (b) $\sqrt{3}$ (c) 2 (d) 4

37. What is $(\sin 9^\circ - \cos 9^\circ)$ equal to?

- (a) $-\frac{\sqrt{5}-\sqrt{5}}{2}$ (b) $-\frac{\sqrt{5}-\sqrt{3}}{2}$
(c) $\frac{\sqrt{5}-\sqrt{5}}{2}$ (d) $\frac{\sqrt{5}-\sqrt{5}}{4}$

38. If in a triangle ABC, $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \sin B \sin C$, then what is the value of the

determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$; where a, b, c are sides of

the triangle?

- (a) $a + b + c$
(b) $ab + bc + ca$
(c) $(a+b)(b+c)(c+a)$
(d) 0

39. If $\cos^{-1} x = \sin^{-1} x$, then which one of the following is correct?

- (a) $x = 1$ (b) $x = \frac{1}{2}$
(c) $x = \frac{1}{\sqrt{2}}$ (d) $x = \frac{1}{\sqrt{3}}$

40. What is the number of solutions of $(\sin \theta - \cos \theta)^2 = 2$ where $-\pi < \theta < \pi$?

- (a) Only one (b) Only two
(c) Four (d) No solution

41. ABC is a triangle such that angle $C = 60^\circ$, then

what is $\frac{\cos A + \cos B}{\cos \left(\frac{A+B}{2} \right)}$ equal to?